



GROUND SOURCE HEAT PUMP RESIDENTIAL AND LIGHT COMMERCIAL **DESIGN AND INSTALLATION GUIDE**

Revision July 3, 2025

This manual is written for professionals or those who wish to become professionals in the Ground Source Heat Pump (GSHP) industry. The International Ground Source Heat Pump Association (IGSHPA) expects the reader to have the appropriate:

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- Knowledge, understanding, and ability to evaluate job site conditions as they apply to your work,
- Equipment and tools necessary to perform your work under job site conditions, and perform same according to the equipment and tool manufacturer's operating manuals and/or recommendations.

Change Log

Date	Page #	Description
03JUL2025	Acknowledge ments	Updated this page to reflect the changes impacting this manual with the publication of CSA/ANSI/IGSHPA C448:25 Series Standard
28JAN2025	4-26	Updated table 4.21
26NOV2021	4-10	Updated Table 4.12
	4-11	Updated Figure 4.1
	4-12	Updated Table 4.13
	4-13	Updated Table 4.14
	4-14	Updated Table 4.15
	4-27	Corrected % methanol from 12.5 to 15
	5-6	Corrected % of antifreezes
	5-7	Updated Figure 5.4
	11-12	Updated Table 11.4

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Ground Source Heat Pump Residential and Light Commercial Design and Installation Guide

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Errors in the manuscript may have occurred in spite of our efforts. Any suggestions, corrections and/or comments should be made in writing to:

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IMPORTANT NOTICE AS OF JULY 3, 2025

The IGSHPA board of directors voted to sunset the 2017 IGSHPA Design and Installation Standards when the next edition of the CSA/ANSI/IGSHPA C448 was published (the edition after the 2016 bi-national edition). The IGSHPA Standards Committee provided feedback to CSA for the next edition (published May 30, 2025). The 2025 edition includes many of the IGSHPA Standards Committee's recommendations and also includes numerous updates from the 2016 edition, which affects the content of this manual. For example, PVC/CPVC piping is no longer listed as acceptable piping material for inside piping of GSHP systems. The new standard also includes new heat transfer fluids such as detoxified ethylene glycol and glycerin-based antifreezes.

The IGSHPA Standards Committee is working on updating this manual as quickly as possible, but it takes a considerable amount of time when major updates are made to the standard. Therefore, this page has been updated to include IGSHPA's position on the recommendations in this manual as compared to the guidelines in CSA/ANSI/IGSHPA C448:25 *Series Design and Installation of Ground Source Heat Pump Systems for Commercial and Residential Buildings*.

NOTICE: All requirements and recommendations in this manual shall be superseded by the guidelines and requirements in CSA/ANSI/IGSHPA C448:25 *Series Design and Installation of Ground Source Heat Pump Systems for Commercial and Residential Buildings* as of the date of this notice. Refer to the C448 Standard design and installation requirements; use this manual as reference, ground heat exchanger design, and other guidelines and best practices not covered by the C448 standard. IGSHPA members may purchase a copy of the standard at <https://igshpa.org/standards>.

$$Q_{\min} = 49.35 \frac{D_{p,i} \cdot \mu}{\rho}$$

Where:

$Q_{\min}$ = Minimum Flow Rate for Turbulence (GPM)

Equation 4.4

Table 4.11. Minimum Flow Rates for Turbulence for Water at 40 F

Pipe Size (in)	ID ¹ (in)	$Q_{\min}$ (gpm)
3/4	0.859	1.05
1	1.076	1.31
1-1/4	1.358	1.66
1-1/2	1.555	1.90
2	1.943	2.38

1. HDPE DR-11 pipe

Adding antifreeze to water to achieve a lower mixture freeze point is common for closed-loop ground heat exchangers in moderate and cold climates. The addition of antifreeze serves to increase the viscosity of the water-antifreeze mixture, especially at low temperatures. Table 4.12 lists the mixture freezing temperature for various volumetric concentrations of propylene glycol, methanol, and ethanol solutions in water. Figure 4.1 displays the freeze protection data graphically.

Table 4.12. Approximate Freeze Protection Levels for Various Types of Antifreeze*

Concentration (by Volume)	Propylene Glycol	Methanol	Ethanol
	Freeze Point (°F)		
5.0%	29.1	27.6	30.0
7.5%	27.6	25.2	28.7
10.0%	26.1	22.7	27.1
12.5%	24.6	20.1	25.3
15.0%	23.0	17.3	23.1
17.5%	21.3	14.4	20.6
20.0%	19.4	11.6	17.9
22.5%	17.4	8.5	14.9
25.0%	15.1	5.3	11.6
27.5%	12.7	—	8.0
30.0%	10.0	—	4.2

*Note - Values are based on pure concentrations of propylene glycol, methanol, and ethanol. Antifreeze products have various levels of inhibitors, dyes, and other ingredients, which will affect all aspects of the antifreeze. Refer to manufacturer's product data for actual values.

Many states require the use of food grade propylene glycol as a freeze protecting agent for closed-loop GSHP systems. The question is how much propylene glycol is actually needed in a given system. The answer, in the case of a GSHP system, is to use at least enough propylene glycol (or any antifreeze) to freeze protect to 10 F below the average loop temperature under design conditions, typically measured on a percent by volume basis. For example, if the minimum design EWT for a loopfield is 30 F, the minimum design LWT will be approxi-

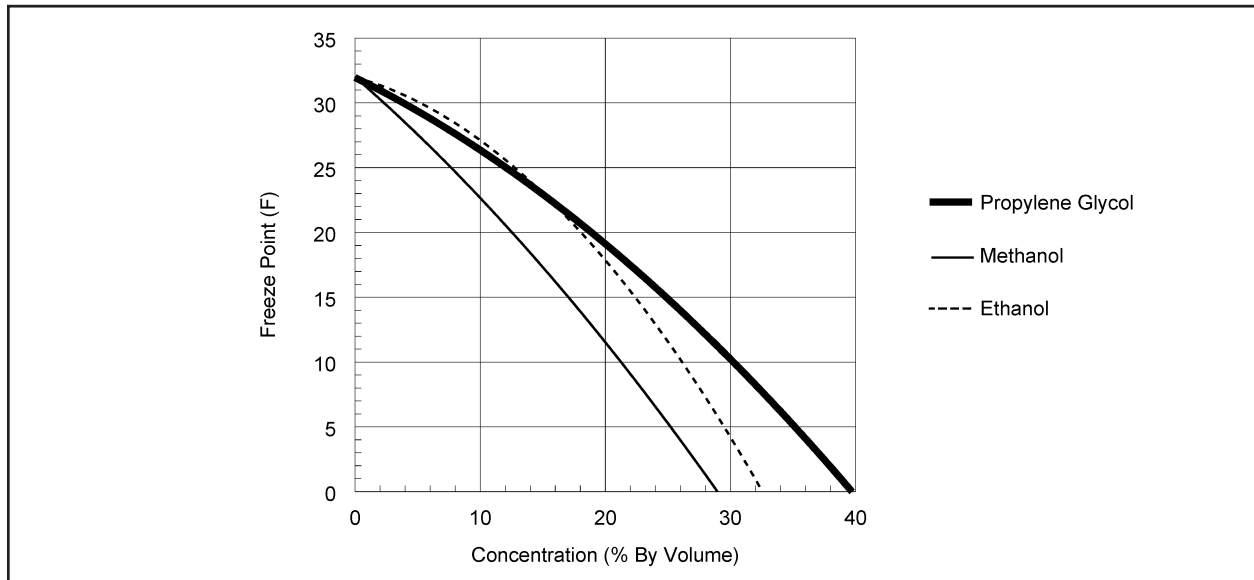


Figure 4.1. Freeze Protection Curves for Various Types of Antifreeze

mately 24-25 F. The average loop temperature under design conditions would be 27 F for this example $[(30+24)/2 = 27 \text{ F}]$. For a 27 F average loop temperature under design conditions, freeze protection to 17 F will be necessary. According to Table 4.12 and Figure 4.1, approximately 20%-22.5% propylene glycol by volume will be necessary for adequate freeze protection for the system. A loopfield designed such that the minimum entering water temperature will not fall below 45 F does not require freeze protection.

Another common question in the industry is whether or not inhibitor content should play a role in the determination of solution concentration level. Inhibitors in propylene glycol-based antifreeze solutions serve three important functions: bacterial growth prevention, corrosion protection, and fluid stability amplification. However, there are very few forms of microbial growth that can actually live without light, and they are relatively rare. The circulating fluid in a closed-loop GSHP system should not be exposed to light in any part of the system, making bacterial growth extremely unlikely and rare. In the event that bacterial growth is found in a system, common water treatment chemicals such as chlorine can be added to impede the growth. Secondly, high inhibitor content for corrosion protection is necessary in systems where the circulating fluid comes into contact with large amounts of corrosive metal. The majority of piping in a typical closed-loop GSHP system is either high-density polyethylene (HDPE) or polyvinyl chloride (PVC). The relatively small amount of metal in a GSHP system that comes into contact with the circulating fluid does not generally necessitate the use of high amounts of corrosion protection. Thirdly, propylene glycol-based solutions will break down when exposed to extreme temperatures greater than 250 F. Ground source heat pump systems operate well below the range of this acceptable temperature; fluid breakdown is a non-issue. For the reasons outlined above, inhibitor content should not play a substantial role in the determination of solution concentration level.

Turbulence for antifreeze-water solutions will occur at a flow rate that depends on the volumetric concentration of antifreeze in the water and the temperature of the mixture. Tables 4.13 through 4.15 provide the minimum flow rate to ensure turbulence ($Re=2,500$) in various HDPE loop diameters for common ranges of propylene glycol, methanol, and ethanol solutions. Fluid viscosities and densities for antifreeze will vary from manufacturer to manufacturer, depending on the type and amount of additives put into the antifreeze mixture to control corrosion, causing slight deviations from the values shown. Refer to manufacturer's data if available.

Table 4.13. Minimum Flow Rates (GPM) for Turbulence with Propylene Glycol

3/4" DR-11 HDPE (ID = 0.859")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	--	--
10%	--	--	--	--	1.87	1.69	1.53
15%	--	--	--	--	2.25	2.03	1.83
20%	--	--	3.54	3.15	2.81	2.52	2.26
25%	--	--	4.73	4.15	3.66	3.23	2.87
30%	8.75	7.56	6.52	5.63	4.88	4.25	3.72
1" DR-11 HDPE (ID = 1.076")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	--	--
10%	--	--	--	--	2.35	2.12	1.92
15%	--	--	--	--	2.82	2.54	2.30
20%	--	--	4.43	3.95	3.52	3.15	2.83
25%	--	--	5.92	5.20	4.58	4.05	3.59
30%	10.96	9.46	8.17	7.06	6.11	5.32	4.66
1-1/4" DR-11 HDPE (ID = 1.358")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	--	--
10%	--	--	--	--	2.96	2.67	2.42
15%	--	--	--	--	3.56	3.21	2.90
20%	--	--	5.59	4.98	4.45	3.98	3.57
25%	--	--	7.47	6.56	5.78	5.11	4.53
30%	13.84	11.95	10.31	8.91	7.72	6.71	5.88
1-1/2" DR-11 HDPE (ID = 1.555")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	--	--
10%	--	--	--	--	3.39	3.06	2.77
15%	--	--	--	--	4.08	3.67	3.32
20%	--	--	6.40	5.71	5.09	4.56	4.09
25%	--	--	8.55	7.52	6.62	5.85	5.19
30%	15.84	13.68	11.81	10.20	8.84	7.69	6.73
2" DR-11 HDPE (ID = 1.943")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	--	--
10%	--	--	--	--	4.24	3.82	3.46
15%	--	--	--	--	5.10	4.59	4.15
20%	--	--	8.00	7.13	6.36	5.69	5.11
25%	--	--	10.69	9.39	8.27	7.30	6.48
30%	19.80	17.09	14.75	12.74	11.04	9.61	8.41

-- Concentration level not recommended due to insufficient freeze protection

Table 4.14. Minimum Flow Rates (GPM) for Turbulence with Methanol

3/4" DR-11 HDPE (ID = 0.859")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	1.15	1.13	1.09
10%	--	--	--	1.94	1.74	1.57	1.42
12.5%	--	--	2.41	2.14	1.91	1.70	1.53
15%	--	--	2.56	2.27	2.01	1.80	1.61
17.5%	--	3.00	2.66	2.36	2.10	1.87	1.68
20%	3.50	3.10	2.75	2.44	2.18	1.95	1.76
1" DR-11 HDPE (ID = 1.076")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	1.44	1.41	1.37
10%	--	--	--	2.43	2.18	1.97	1.78
12.5%	--	--	3.02	2.68	2.39	2.13	1.91
15%	--	--	3.21	2.84	2.52	2.25	2.02
17.5%	--	3.76	3.33	2.95	2.63	2.35	2.11
20%	4.38	3.88	3.44	3.06	2.73	2.44	2.20
1-1/4" DR-11 HDPE (ID = 1.358")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	1.81	1.78	1.73
10%	--	--	--	3.06	2.75	2.48	2.25
12.5%	--	--	3.81	3.38	3.01	2.69	2.41
15%	--	--	4.05	3.59	3.19	2.84	2.55
17.5%	--	4.75	4.20	3.72	3.31	2.96	2.66
20%	5.53	4.90	4.34	3.86	3.44	3.08	2.78
1-1/2" DR-11 HDPE (ID = 1.555")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	2.08	2.04	1.98
10%	--	--	--	3.51	3.15	2.84	2.57
12.5%	--	--	4.36	3.88	3.45	3.08	2.76
15%	--	--	4.64	4.11	3.65	3.25	2.92
17.5%	--	5.44	4.81	4.26	3.79	3.39	3.05
20%	6.33	5.61	4.97	4.42	3.94	3.53	3.18
2" DR-11 HDPE (ID = 1.943")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	2.59	2.55	2.47
10%	--	--	--	4.38	3.94	3.55	3.21
12.5%	--	--	5.45	4.84	4.31	3.85	3.45
15%	--	--	5.79	5.13	4.56	4.06	3.64
17.5%	--	6.79	6.01	5.33	4.74	4.24	3.81
20%	7.91	7.00	6.21	5.52	4.92	4.41	3.98

-- Concentration level not recommended due to insufficient freeze protection

Table 4.15. Minimum Flow Rates (GPM) for Turbulence with Ethanol

3/4" DR-11 HDPE (ID = 0.859")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	2.63	2.24
10%	--	--	--	--	2.74	2.44	2.17
15%	--	--	--	3.22	2.90	2.61	2.35
20%	--	--	4.14	3.72	3.34	3.00	2.69
25%	6.08	5.43	4.84	4.31	3.85	3.43	3.07
30%	6.69	5.96	5.31	4.73	4.22	3.76	3.36
1" DR-11 HDPE (ID = 1.076")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	3.29	2.80
10%	--	--	--	--	3.43	3.05	2.71
15%	--	--	--	4.03	3.63	3.27	2.95
20%	--	--	5.19	4.66	4.18	3.75	3.37
25%	7.62	6.80	6.06	5.41	4.82	4.30	3.84
30%	8.38	7.47	6.65	5.93	5.28	4.71	4.21
1-1/4" DR-11 HDPE (ID = 1.358")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	4.15	3.54
10%	--	--	--	--	4.33	3.85	3.42
15%	--	--	--	5.09	4.58	4.13	3.72
20%	--	--	6.55	5.88	5.28	4.74	4.25
25%	9.61	8.58	7.65	6.82	6.08	5.43	4.85
30%	10.57	9.43	8.40	7.48	6.66	5.95	5.32
1-1/2" DR-11 HDPE (ID = 1.555")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	4.75	4.05
10%	--	--	--	--	4.96	4.41	3.92
15%	--	--	--	5.82	5.25	4.73	4.26
20%	--	--	7.50	6.73	6.04	5.42	4.87
25%	11.01	9.83	8.76	7.81	6.96	6.21	5.55
30%	12.11	10.79	9.61	8.56	7.63	6.81	6.09
2" DR-11 HDPE (ID = 1.943")							
Concentration (By Volume)	Fluid Temperature (°F)						
	10 °F	15 °F	20 °F	25 °F	30 °F	35 °F	40 °F
5%	--	--	--	--	--	5.94	5.06
10%	--	--	--	--	6.20	5.51	4.90
15%	--	--	--	7.28	6.56	5.91	5.32
20%	--	--	9.37	8.41	7.55	6.78	6.09
25%	13.76	12.28	10.95	9.76	8.70	7.76	6.94
30%	15.13	13.49	12.01	10.70	9.54	8.51	7.61

-- Concentration level not recommended due to insufficient freeze protection

4.4.4 ANTIFREEZE CORRECTION FACTORS

Because water-antifreeze solutions tend to be more viscous (and therefore harder to pump) than water itself, correction factors are needed to account for such viscosity increase for pump sizing purposes. In many cases, manufacturers will provide head loss tables based on water at 40 F as the circulating fluid. Antifreeze correction factors are calculated by dividing the head loss caused by the antifreeze at the temperature and concentration of interest by the head loss caused by water at 40 F. Again, keep in mind that fluid viscosities and densities for antifreeze will vary from manufacturer to manufacturer. Table 4.21 provides antifreeze viscosity correction factors for various types and concentrations of antifreeze at various temperatures.

Table 4.21. Antifreeze Viscosity Correction Factors (C_{VAF})

Propylene Glycol Viscosity Multipliers								
Concentration By Volume	Freeze Point (°F)	Temperature (°F)						
		40	35	30	25	20	15	10
10%	26.1	1.13	1.16	1.19	--	--	--	--
15%	23.0	1.19	1.22	1.25	--	--	--	--
20%	19.4	1.24	1.28	1.31	1.35	1.39	--	--
25%	15.1	1.34	1.38	1.42	1.47	1.52	--	--
30%	10.0	1.41	1.46	1.51	1.56	1.62	1.68	1.74
Methanol Viscosity Multipliers								
Concentration By Volume	Freeze Point (°F)	Temperature (°F)						
		40	35	30	25	20	15	10
5%	27.6	1.08	1.11	--	--	--	--	--
10%	22.7	1.13	1.15	1.18	1.22	--	--	--
12.5%	20.1	1.15	1.18	1.21	1.25	--	--	--
15%	17.3	1.17	1.20	1.23	1.27	1.31	1.34	--
17.5%	14.4	1.18	1.21	1.25	1.29	1.33	1.36	--
20%	11.6	1.20	1.23	1.26	1.31	1.35	1.38	--
Ethanol Viscosity Multipliers								
Concentration By Volume	Freeze Point (°F)	Temperature (°F)						
		40	35	30	25	20	15	10
10%	27.1	1.21	1.26	1.30	--	--	--	--
15%	23.1	1.26	1.31	1.36	1.40	--	--	--
20%	17.9	1.31	1.36	1.42	1.47	1.56	--	--
25%	11.6	1.35	1.41	1.47	1.57	1.64	1.67	--
30%	4.2	1.37	1.44	1.53	1.63	1.66	1.69	--

-- Concentration level not recommended due to insufficient freeze protection

Use Equation 4.8 to calculate head loss in a given type/size of pipe for antifreeze solutions of different temperatures/concentrations.

$$HL_{\text{AF}} = HL_{\text{H}_2\text{O}} \cdot C_{\text{VAF}}$$

Equation 4.8

Where:

HL_{AF} = Head Loss for a Given Antifreeze Solution Type/Concentration/Temp (ft)

$HL_{\text{H}_2\text{O}}$ = Head Loss for Water at 40 F (ft - refer to Tables 4.16-4.20 and Figures 4.2-4.8)

C_{VAF} = Viscosity Correction Factor for Given Antifreeze Solution Type/Concentration/Temp (refer to Table 4.21)

4.5 Flow Capacities

When selecting the size of pipe to be used as the ground heat exchanger portion of the GSHP system, several guidelines must be followed. First, pipe sizing needs to be large enough to keep pumping power requirements to a minimum. Secondly, pipe sizing needs to be small enough to ensure turbulent flow in the circulating fluid, thus maximizing heat transfer between the fluid and the pipe wall. A simple guideline to follow is to size the pipe such that the required flow will result in 1-3 feet of head loss per 100 feet of pipe for any type of circulating fluid. Designing the ground heat exchanger to stay within this range will ensure the proper balance between minimizing pumping power requirements and material costs and maximizing thermal performance. Pipe sizing that produces head loss significantly below this range will cause the pipe to be oversized and can dramatically increase project material and handling costs in addition to allowing fluid flow to fall out of the turbulent regime in cases where a water and antifreeze solution is used. Pipe sizing that produces head loss above this range can cause significantly higher system first cost and operating cost due to increased pump size and pumping power demand. Table 4.22 provides flow capacities for two common HDPE DRs and a range of pipe sizes to stay within the range of 1-3 feet of head loss per 100 feet of pipe for water at 40 F.

Table 4.22. Flow Capacities for Various Pipe Sizes for 1-3 feet of Head Loss / 100 feet of Pipe (Water at 40 F)

DR-11		DR-15.5	
Nom. Size (in)	Capacity (gpm)	Nom. Size (in)	Capacity (gpm)
3/4	2.0 - 3.7	2	22.5 - 41.7
1	3.7 - 7.3	3	63.5 - 117
1-1/4	7.3 - 13.6	4	124 - 229
1-1/2	10.5 - 19.5	6	348 - 640
2	19.1 - 35.3	8	700 - 1290
3	53.8 - 99.4	10	1260 - 2300

In some instances, it is not possible to stay within the recommended range of 1-3 feet of head loss per 100 feet of pipe length. The maximum recommended pressure drop for a closed-loop system is 4 feet of head loss per 100 feet of pipe. Tables 4.23 through 4.25 provide the maximum recommended fluid flow rates for various types and sizes of pipe. The maximum recommended flow rates given in the tables are based on 4 feet of head per 100 feet of pipe for water at 40 F, 20% propylene glycol by volume at 25 F, 15% methanol by volume at 25 F, and 20% ethanol by volume at 25 F.

flow rate through each GHEX flow path is required for turbulent flow, and a maximum flow rate exists where the head loss becomes excessive (approximately 4 feet of head loss per 100 feet of pipe). For a given pipe size, the acceptable flow rate range depends on the type and temperature of the circulating fluid. Minimum flow rates for turbulence in DR-11 HDPE pipe are provided in Table 4.10 for water at 40 F and in Tables 4.12 through 4.14 for propylene glycol, methanol, and ethanol solutions, respectively, at various concentrations and temperatures. Maximum flow rates, resulting in 4 feet of head loss per 100 feet of pipe, can be estimated from Table 4.15 for water at 40 F and for 20% propylene glycol, 15% methanol, and 20% ethanol at 25 F. These percentages provide similar freeze points of 19.4 F, 17.3 F, and 17.9 F, respectively, for the three antifreeze solutions and represent the appropriate freeze protection level of about 10 F below the average circulating fluid temperature for a design minimum entering water temperature of 30 F, where the lowest temperature in the GHEX (where the fluid is most viscous) would be about 25 F.

Figure 5.4 summarizes the minimum and maximum flow rates for the four circulating fluids discussed above. Water allows the largest flow rate range for all pipe sizes, followed by methanol, propylene glycol, and ethanol. For all circulating fluids, $\frac{3}{4}$ -inch DR-11 provides the narrowest acceptable flow rate range, with ethanol having virtually no difference between minimum and maximum acceptable flow rates. Minimum flow rates for turbulence and maximum flow rates for head loss can be determined for other antifreeze solutions and design temperatures from the data in Chapter 4.

Figure 5.4 provides a starting point for pipe size selection and the number of flow paths in the GHEX. In residential and light commercial design, the general rule is to design for one flow path per nominal ton of heat pump capacity. Because 1 ton of heat pump capacity requires approximately 3 GPM of flow, one flow path would accommodate that same 3 GPM of flow. Looking at Figure 5.4, only the flow rate range associated with $\frac{3}{4}$ -inch pipe (approximately 1.00 - 4.75 GPM for water, 2.25 - 4.00 for methanol, 3.25 - 3.75 for propylene glycol, and 3.65 for ethanol) would meet both the minimum and maximum flow rate constraints for all circulating fluids (very close for ethanol). To achieve turbulence at 3 GPM, both water and methanol would work for 1-inch pipe and only water for 1-1/4-inch pipe. In vertically and horizontally-bored GHEX systems, depending on drilling conditions and capabilities, the design bore length may frequently represent more than 1 nominal ton of heat pump capacity (Figure 5.5), and the required flow rate through each ground loop may require 1-inch or 1-1/4-inch pipe to control head loss. Looking at Figure 5.4, the 1-inch pipe should accommodate between 4.5 and 6 GPM for most circulating fluids and the 1-1/4-inch pipe between 6 and 12 GPM, representing 1.5 to 2.0 tons and 2.0 to 4.0 tons of heat pump capacity for the 1-inch and 1-1/4-inch pipes, respectively. Looking at Figure 5.5 and considering constant soil or rock thermal properties with depth, a 3-ton GHEX may be comprised of three bores to a given design depth (D_3) using $\frac{3}{4}$ -inch pipe, two bores to a depth D_2 approximately 1.5 times deeper than D_3 using 1-inch pipe, or one bore approximately 3 times deeper ($D_1 \approx 3D_3$) using 1-1/4-inch pipe. Of course, if the soil or rock thermal properties change with depth, the relative design lengths of the three configurations in Figure 5.5 will change.

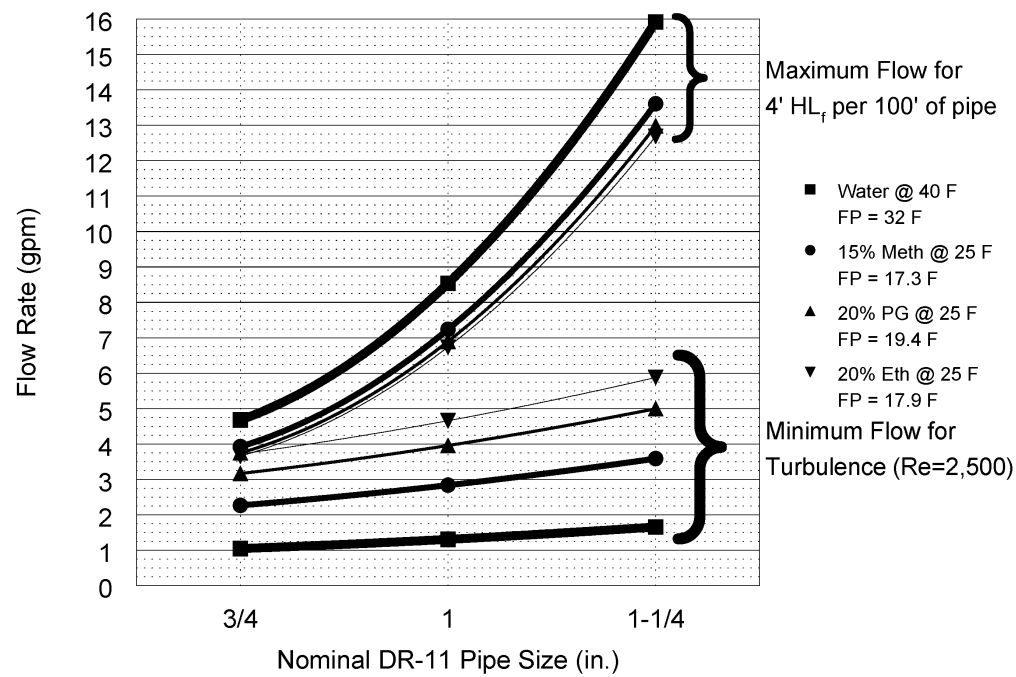


Figure 5.4. Flow rate range bands for selected circulating fluids.

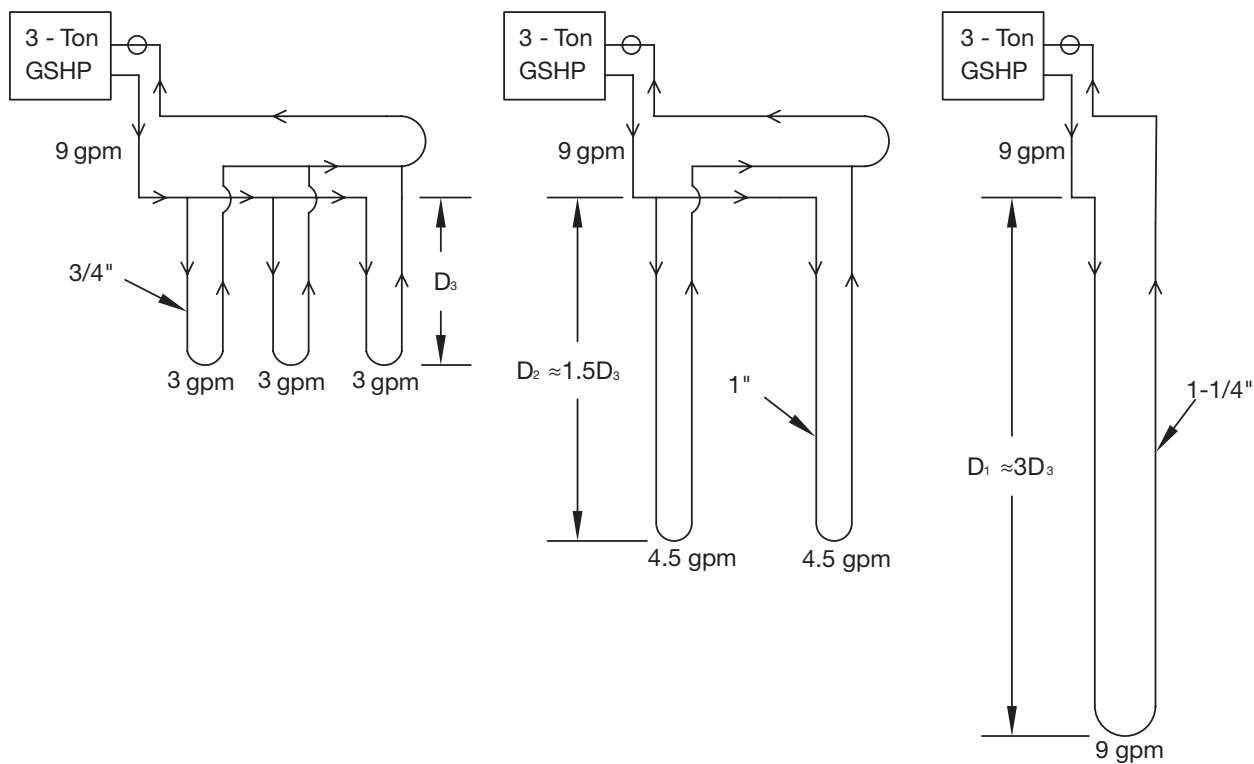


Figure 5.5. Optional ground loop layouts for a nominal “3-ton” GSHP.

Table 11.4. Flow Constants for Various Antifreeze Solutions at Selected Temperatures

Flow Constants (K) for Propylene Glycol						
Concentration By Volume	Freeze Point (°F)	Temperature (°F)				
		30	35	40	45	50
10%	26.1	491	491	491	492	492
15%	23.0	487	488	488	488	488
20%	19.4	483	483	483	484	484
25%	15.1	477	477	478	478	479
30%	10.0	471	471	472	472	472
Flow Constants (K) for Methanol						
Concentration By Volume	Freeze Point (°F)	Temperature (°F)				
		30	35	40	45	50
5%	27.6	520	519	519	519	519
10%	22.7	508	508	507	507	507
12.5%	20.1	502	502	501	501	501
15%	17.3	497	496	496	496	495
17.5%	14.4	491	491	490	490	490
20%	11.6	486	485	485	485	484
Flow Constants (K) for Ethanol						
Concentration By Volume	Freeze Point (°F)	Temperature (°F)				
		30	35	40	45	50
10%	27.1	524	523	523	522	521
15%	23.1	521	521	520	520	519
20%	17.9	516	515	514	514	513
25%	11.6	507	506	505	505	504
30%	4.2	494	494	493	492	491

Electric demand (DMD) is the rate at which the GSHP consumes electrical power and its calculation depends on whether the electrical service is single or three-phase, as described by Equation 11.3a (single-phase) and Equation 11.3b (three-phase) (ECONAR Energy Systems).

Single-Phase:

$$\text{DMD} = 0.85 \cdot V \cdot I$$

Equation 11.3a

Three-Phase:

$$\text{DMD} = 0.85 \cdot 1.732 \cdot V \cdot I = 1.472 \cdot V \cdot I$$

Equation 11.3b

Where:

DMD = Electric demand (W)

V = Voltage supplied to GSHP (V)

I = Amperage draw through GSHP (A)

0.85 = Power factor for inductive loads, assumed average value

1.732 = Line voltage = $2 \cdot (\text{Phase voltage}) \cdot \sin(120^\circ) = \sqrt{3}$

Coefficient of Performance (COP) is a measure of the GSHP efficiency in the heating mode and is calculated using Equation 11.4.